

## 1 v0.8 벡터 조립 도면 세트

물리 검증·안전 인증·통전 승인이 아니다. E-stop, lid/service interlock, branch fuse, 독립 thermal fuse를 정상 firmware와 독립 구현하고 exact donor 정격·배선·보호소자를 실측 확인하기 전 통전하지 않는다.

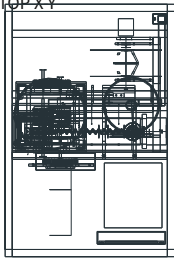
Revision: final-design-fabrication-closure-v0.8 · 상태: DIGITAL\_DOCUMENT / PHYSICAL\_NOT\_RUN / USER\_APPROVAL\_REQUIRED

## 2 GA-001 — general arrangement

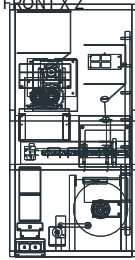
### GA-001 general arrangement

v0.8 · mm · 제3각법 · source: FreeCAD Python · bodies 171

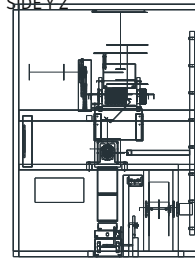
TOP X-Y



FRONT X-Z



SIDE Y-Z



Overall X 470.00 · Y 700.00 · Z 930.00 mm · 치수/공차는 개별 제작도면과 interface catalog 우선  
Revision final-design-fabrication-closure-v0.8 · physical validation NOT\_RUN

Drawing GA-001 · Rev v0.8 · mm · third-angle · NTS · source 65f758c4b46b520eafd86cf3667ed249af2fb5f2

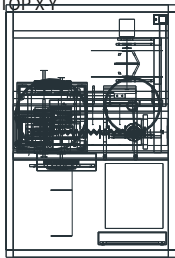
General tolerance ISO 2768-m. Critical interfaces are controlled by exports/final/interface\_catalog.csv; do not scale this drawing.

### 3 ASM-001 — full assembly

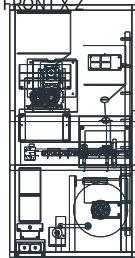
#### ASM-001 full assembly

v0.8 · mm · 제3각법 · source: FreeCAD Python · bodies 171

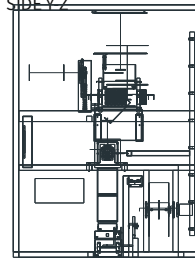
TOP X-Y



FRONT X-Z



SIDE Y-Z



Overall X 470.00 · Y 700.00 · Z 930.00 mm · 치수/공차는 개별 제작도면과 interface catalog 우선  
Revision final-design-fabrication-closure-v0.8 · physical validation NOT\_RUN

Drawing ASM-001 · Rev v0.8 · mm · third-angle · NTS · source 65f758c4b46b520eafd86cf3667ed249af2fb5f2

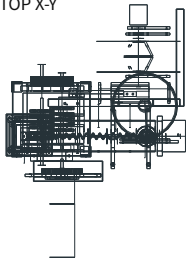
General tolerance ISO 2768-m. Critical interfaces are controlled by `exports/final/interface_catalog.csv`; do not scale this drawing.

## 4 ASM-002 — module arrangement

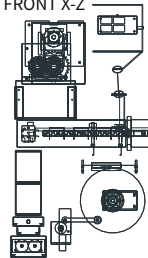
### ASM-002 module arrangement

v0.8 · mm · 제3각법 · source: FreeCAD Python · bodies 128

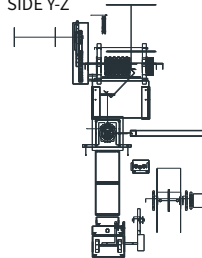
TOP X-Y



FRONT X-Z



SIDE Y-Z



Overall X 427.00 · Y 601.00 · Z 780.00 mm · 치수/공차는 개별 제작도면과 interface catalog 우선

Revision final-design-fabrication-closure-v0.8 · physical validation NOT\_RUN

Drawing ASM-002 · Rev v0.8 · mm · third-angle · NTS · source 65f758c4b46b520eafd86cf3667ed249af2fb5f2

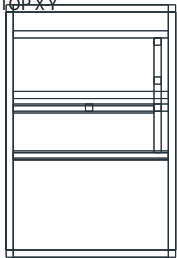
General tolerance ISO 2768-m. Critical interfaces are controlled by `exports/final/interface_catalog.csv`; do not scale this drawing.

## 5 FR-001 — frame

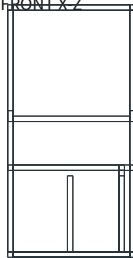
### FR-001 frame

v0.8 · mm · 제3각법 · source: FreeCAD Python · bodies 28

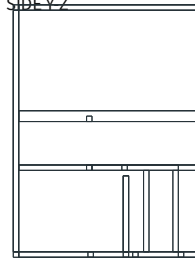
TOP X-Y



FRONT X-Z



SIDE Y-Z

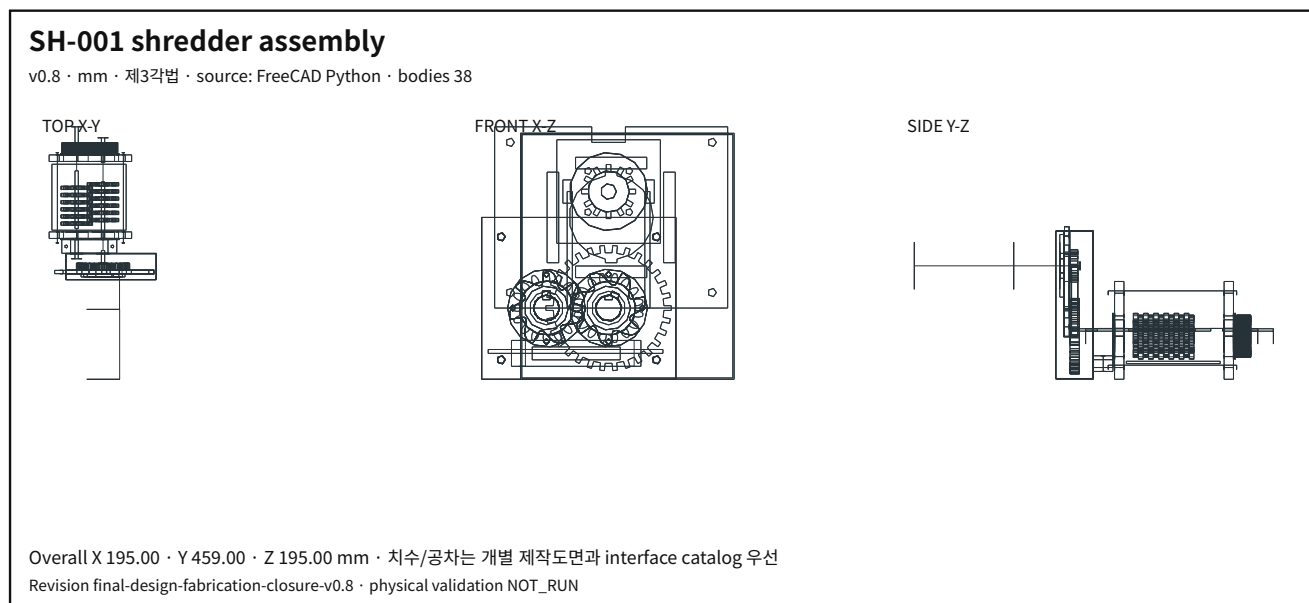


Overall X 470.00 · Y 700.00 · Z 930.00 mm · 치수/공차는 개별 제작도면과 interface catalog 우선  
Revision final-design-fabrication-closure-v0.8 · physical validation NOT\_RUN

Drawing FR-001 · Rev v0.8 · mm · third-angle · NTS · source 65f758c4b46b520eafd86cf3667ed249af2fb5f2

General tolerance ISO 2768-m. Critical interfaces are controlled by exports/final/interface\_catalog.csv; do not scale this drawing.

## 6 SH-001 — shredder assembly



Drawing SH-001 · Rev v0.8 · mm · third-angle · NTS · source 65f758c4b46b520eafd86cf3667ed249af2fb5f2

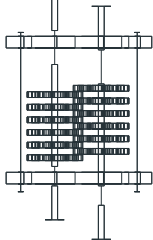
General tolerance ISO 2768-m. Critical interfaces are controlled by `exports/final/interface_catalog.csv`; do not scale this drawing.

## 7 SH-002 — cutter stack

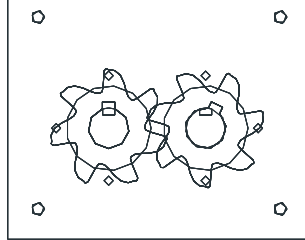
### SH-002 cutter stack

v0.8 · mm · 제3각법 · source: FreeCAD Python · bodies 20

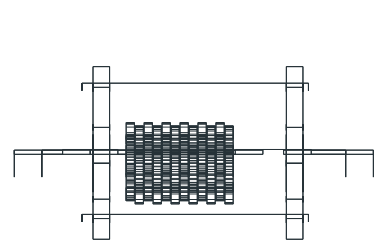
TOP X-Y



FRONT X-Z



SIDE Y-Z



Overall X 150.00 · Y 260.00 · Z 125.00 mm · 치수/공차는 개별 제작도면과 interface catalog 우선  
Revision final-design-fabrication-closure-v0.8 · physical validation NOT\_RUN

Drawing SH-002 · Rev v0.8 · mm · third-angle · NTS · source 65f758c4b46b520eafd86cf3667ed249af2fb5f2

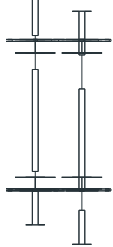
General tolerance ISO 2768-m. Critical interfaces are controlled by `exports/final/interface_catalog.csv`; do not scale this drawing.

## 8 SH-003 — shaft and bearing assembly

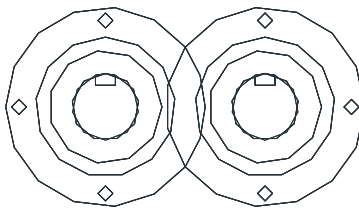
### SH-003 shaft bearing

v0.8 · mm · 제3각법 · source: FreeCAD Python · bodies 8

TOP X-Y



FRONT X-Z



SIDE Y-Z

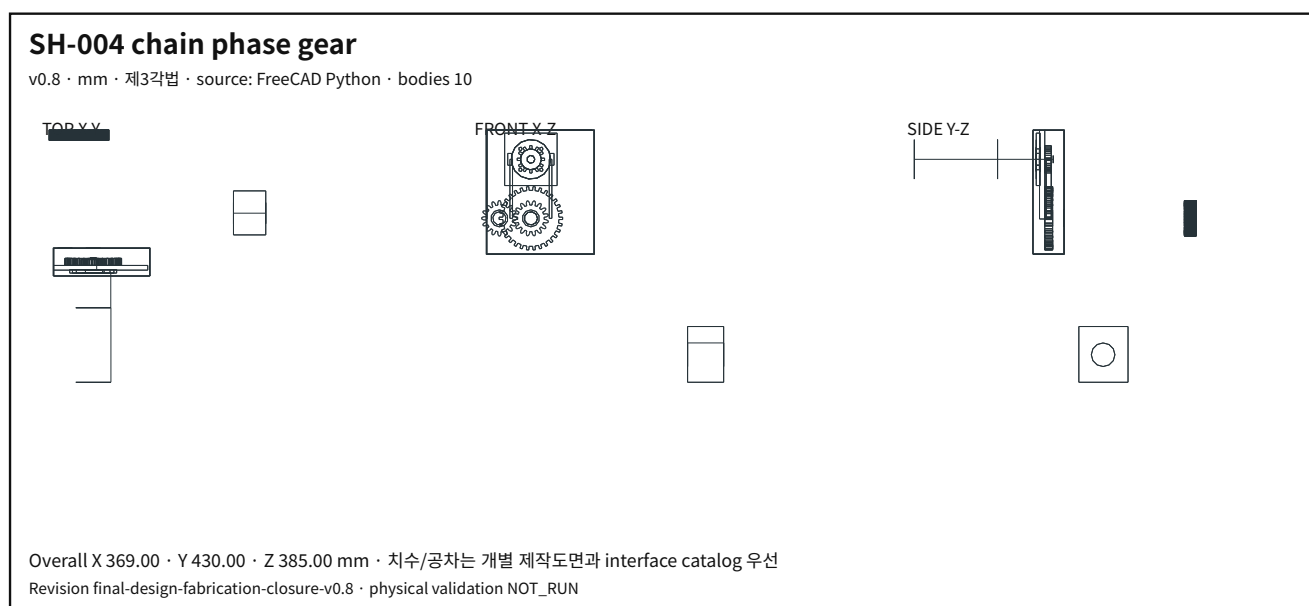


Overall X 108.00 · Y 260.00 · Z 60.00 mm · 치수/공차는 개별 제작도면과 interface catalog 우선  
Revision final-design-fabrication-closure-v0.8 · physical validation NOT\_RUN

Drawing SH-003 · Rev v0.8 · mm · third-angle · NTS · source 65f758c4b46b520eafd86cf3667ed249af2fb5f2

General tolerance ISO 2768-m. Critical interfaces are controlled by `exports/final/interface_catalog.csv`; do not scale this drawing.

## 9 SH-004 — chain and phase gear



Drawing SH-004 · Rev v0.8 · mm · third-angle · NTS · source 65f758c4b46b520eafd86cf3667ed249af2fb5f2

General tolerance ISO 2768-m. Critical interfaces are controlled by `exports/final/interface_catalog.csv`; do not scale this drawing.

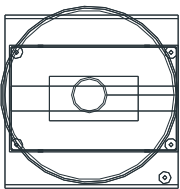


## 10 FD-001 — hopper

### FD-001 hopper

v0.8 · mm · 제3각법 · source: FreeCAD Python · bodies 11

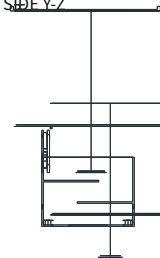
TOP X-Y



FRONT X-Z



SIDE Y-Z

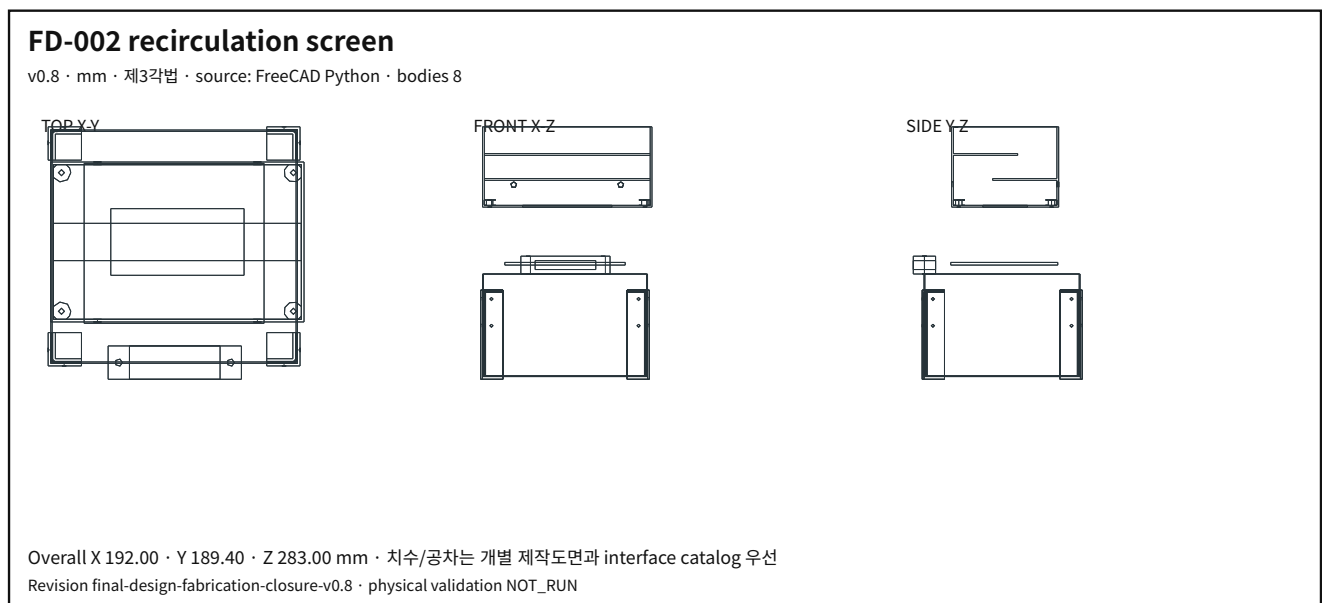


Overall X 405.00 · Y 208.00 · Z 328.00 mm · 치수/공차는 개별 제작도면과 interface catalog 우선  
Revision final-design-fabrication-closure-v0.8 · physical validation NOT\_RUN

Drawing FD-001 · Rev v0.8 · mm · third-angle · NTS · source 65f758c4b46b520eafd86cf3667ed249af2fb5f2

General tolerance ISO 2768-m. Critical interfaces are controlled by exports/final/interface\_catalog.csv; do not scale this drawing.

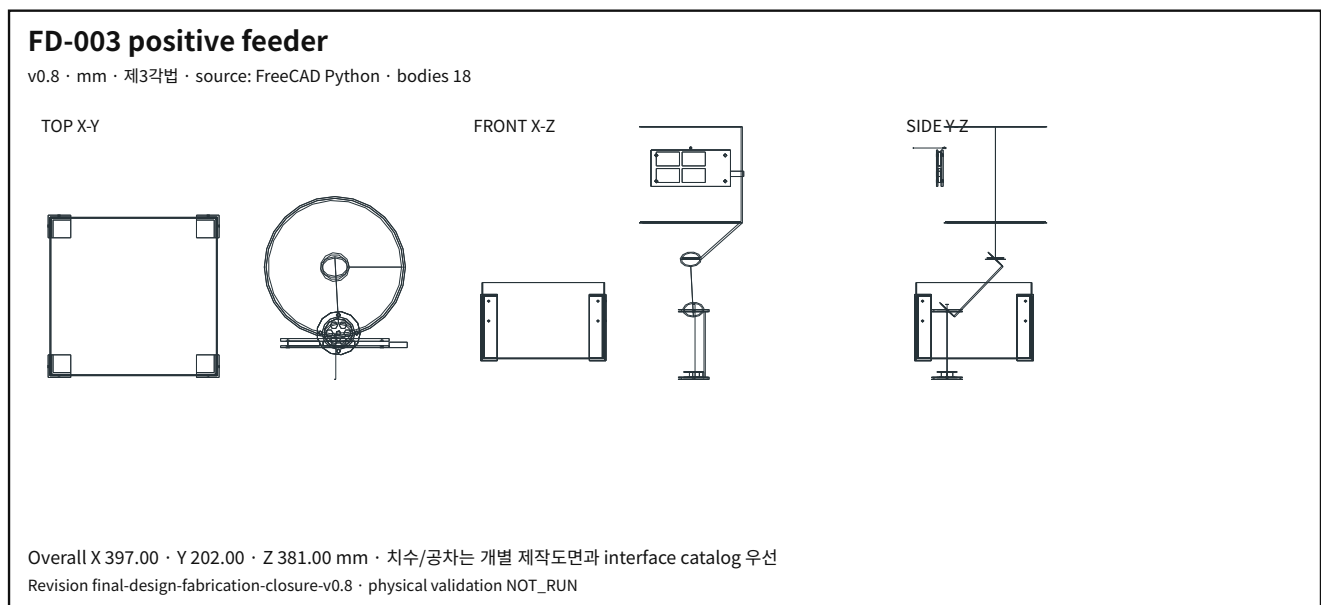
## 11 FD-002 — recirculation/screen



Drawing FD-002 · Rev v0.8 · mm · third-angle · NTS · source 65f758c4b46b520eafd86cf3667ed249af2fb5f2

General tolerance ISO 2768-m. Critical interfaces are controlled by `exports/final/interface_catalog.csv`; do not scale this drawing.

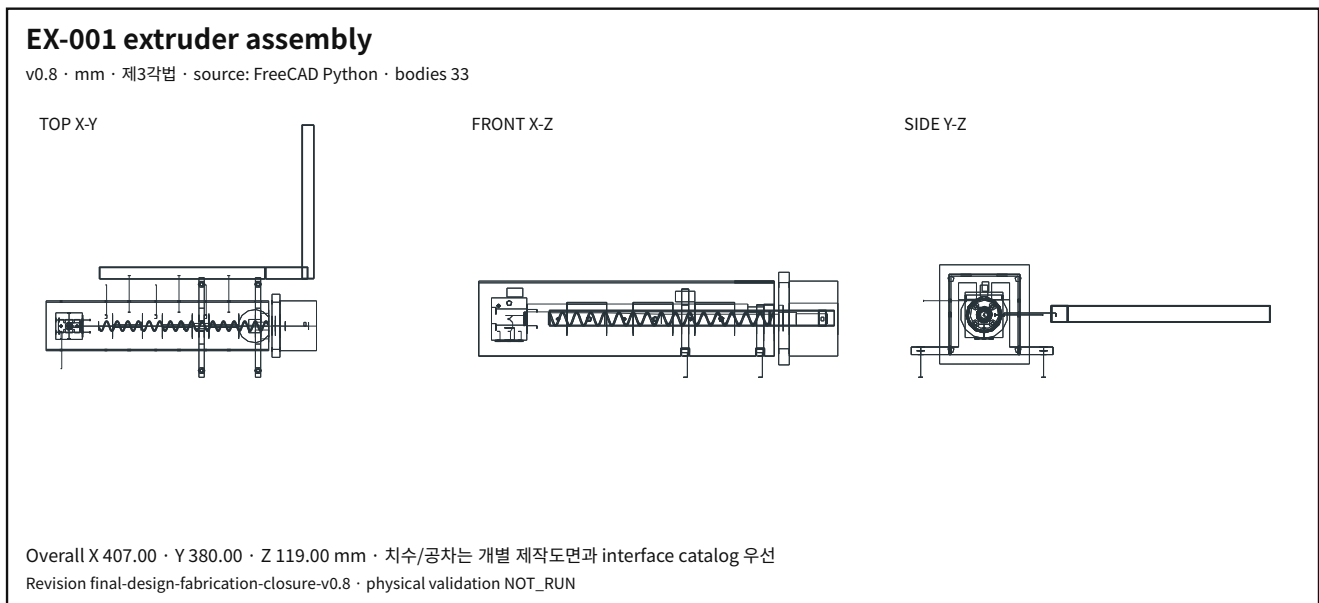
## 12 FD-003 — positive feeder



Drawing FD-003 · Rev v0.8 · mm · third-angle · NTS · source 65f758c4b46b520eafd86cf3667ed249af2fb5f2

General tolerance ISO 2768-m. Critical interfaces are controlled by `exports/final/interface_catalog.csv`; do not scale this drawing.

## 13 EX-001 — extruder assembly



Drawing EX-001 · Rev v0.8 · mm · third-angle · NTS · source 65f758c4b46b520eafd86cf3667ed249af2fb5f2

General tolerance ISO 2768-m. Critical interfaces are controlled by `exports/final/interface_catalog.csv`; do not scale this drawing.

## 14 EX-002 — screw/barrel/die

### EX-002 screw barrel die

v0.8 · mm · 제3각법 · source: FreeCAD Python · bodies 13

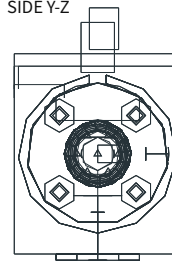
TOP X-Y



FRONT X-Z



SIDE Y-Z



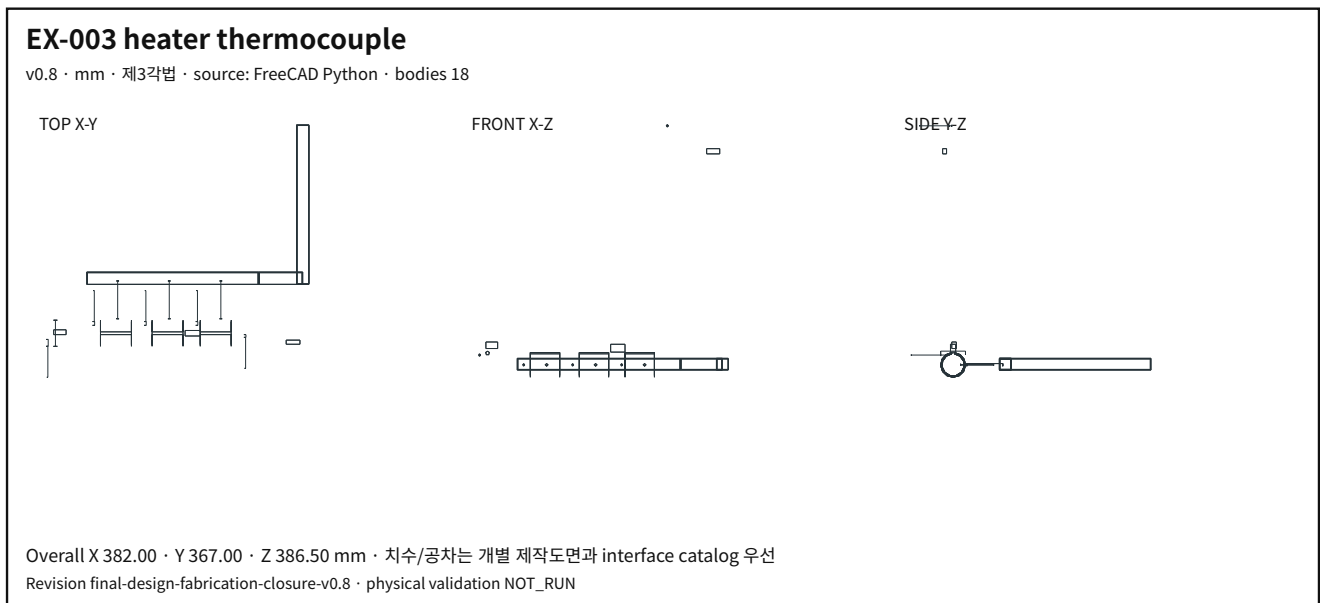
Overall X 380.50 · Y 40.00 · Z 60.50 mm · 치수/공차는 개별 제작도면과 interface catalog 우선

Revision final-design-fabrication-closure-v0.8 · physical validation NOT\_RUN

Drawing EX-002 · Rev v0.8 · mm · third-angle · NTS · source 65f758c4b46b520eafd86cf3667ed249af2fb5f2

General tolerance ISO 2768-m. Critical interfaces are controlled by `exports/final/interface_catalog.csv`; do not scale this drawing.

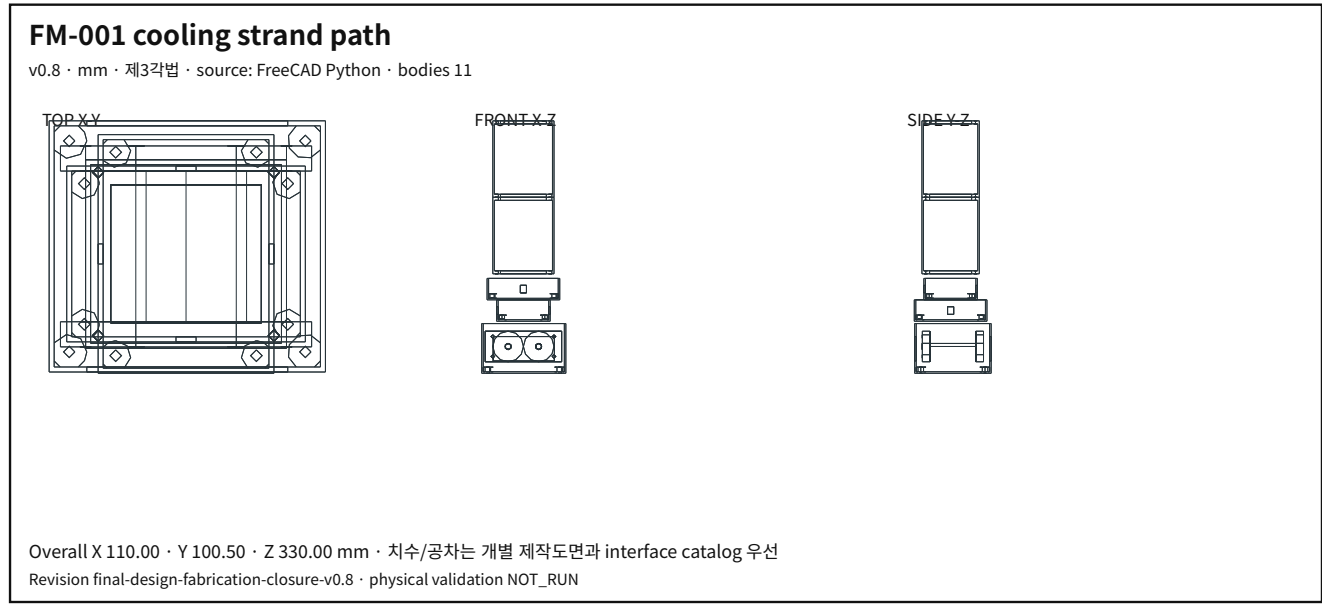
## 15 EX-003 — heater/thermocouple layout



Drawing EX-003 · Rev v0.8 · mm · third-angle · NTS · source 65f758c4b46b520eafd86cf3667ed249af2fb5f2

General tolerance ISO 2768-m. Critical interfaces are controlled by `exports/final/interface_catalog.csv`; do not scale this drawing.

16 FM-001 — cooling and strand path

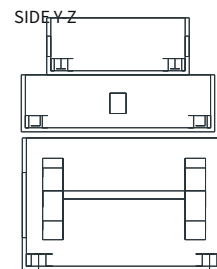
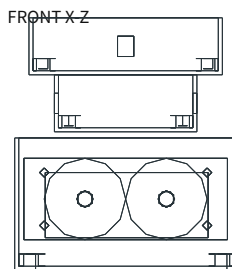
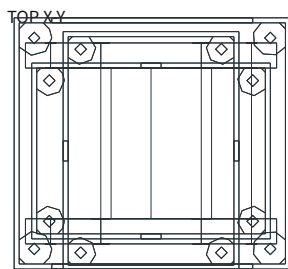


Drawing FM-001 · Rev v0.8 · mm · third-angle · NTS · source 65f758c4b46b520eafd86cf3667ed249af2fb5f2  
General tolerance ISO 2768-m. Critical interfaces are controlled by exports/final/interface\_catalog.csv; do not scale this drawing.

## 17 FM-002 — gauge/puller

### FM-002 gauge puller

v0.8 · mm · 제3각법 · source: FreeCAD Python · bodies 9



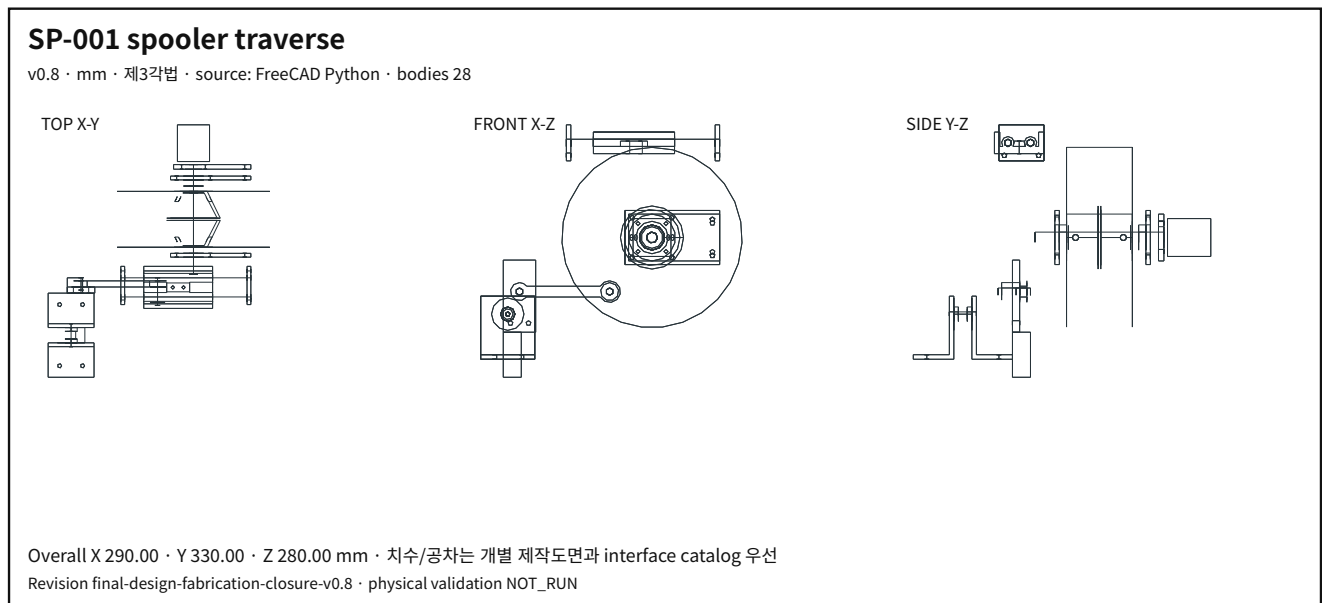
Overall X 110.00 · Y 100.50 · Z 124.00 mm · 치수/공차는 개별 제작도면과 interface catalog 우선  
Revision final-design-fabrication-closure-v0.8 · physical validation NOT\_RUN

Drawing FM-002 · Rev v0.8 · mm · third-angle · NTS · source 65f758c4b46b520eafd86cf3667ed249af2fb5f2

General tolerance ISO 2768-m. Critical interfaces are controlled by `exports/final/interface_catalog.csv`; do not scale this drawing.



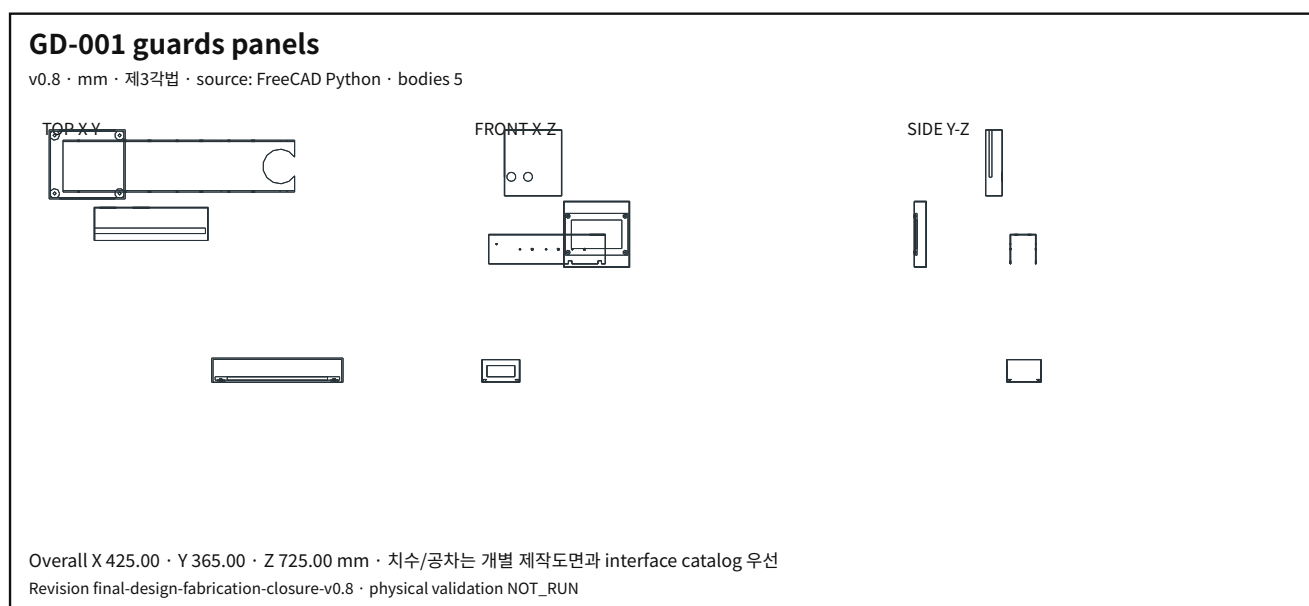
## 18 SP-001 — spooler/traverse



Drawing SP-001 · Rev v0.8 · mm · third-angle · NTS · source 65f758c4b46b520eafd86cf3667ed249af2fb5f2

General tolerance ISO 2768-m. Critical interfaces are controlled by `exports/final/interface_catalog.csv`; do not scale this drawing.

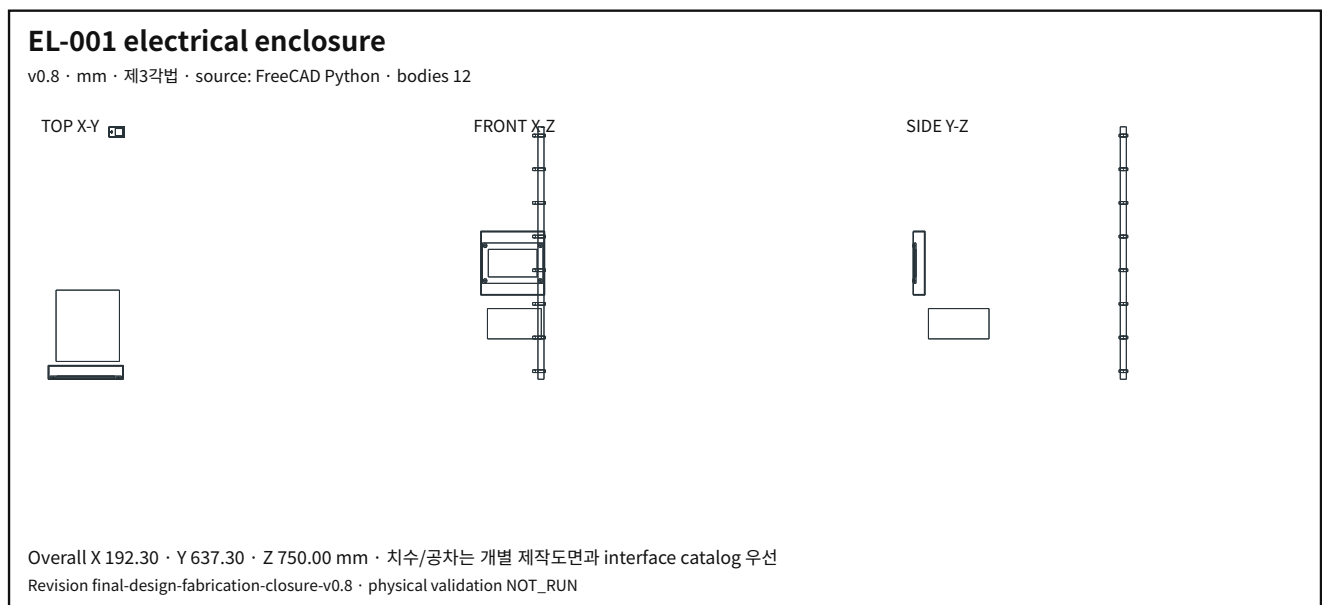
## 19 GD-001 — guards and panels



Drawing GD-001 · Rev v0.8 · mm · third-angle · NTS · source 65f758c4b46b520eafd86cf3667ed249af2fb5f2

General tolerance ISO 2768-m. Critical interfaces are controlled by `exports/final/interface_catalog.csv`; do not scale this drawing.

## 20 EL-001 — electrical enclosure



Drawing EL-001 · Rev v0.8 · mm · third-angle · NTS · source 65f758c4b46b520eafd86cf3667ed249af2fb5f2

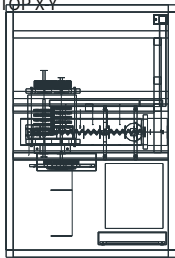
General tolerance ISO 2768-m. Critical interfaces are controlled by `exports/final/interface_catalog.csv`; do not scale this drawing.

## 21 SV-001 — service envelopes

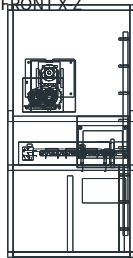
### SV-001 service envelopes

v0.8 · mm · 제3각법 · source: FreeCAD Python · bodies 111

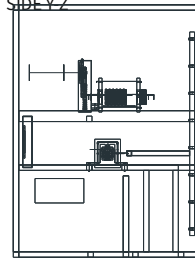
TOP X-Y



FRONT X-Z



SIDE Y-Z



Overall X 470.00 · Y 700.00 · Z 930.00 mm · 치수/공차는 개별 제작도면과 interface catalog 우선  
Revision final-design-fabrication-closure-v0.8 · physical validation NOT\_RUN

Drawing sv-001 · Rev v0.8 · mm · third-angle · NTS · source 65f758c4b46b520eafd86cf3667ed249af2fb5f2

General tolerance ISO 2768-m. Critical interfaces are controlled by `exports/final/interface_catalog.csv`; do not scale this drawing.